Ryan DeBraal

Principal Full-Stack Software Engineer · CEO, MindAttic LLC · President, MindAttic Cares

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SUMMARY

Principal-level full-stack engineer with 20+ years architecting high-impact systems for Fortune 500 organizations. Inventor with three patents across UI innovation, security platforms, and data visualization. Depth across the entire stack — from pixel-perfect UI/UX and component library design to distributed backend architecture, multi-threaded data engines, and machine learning pipelines. Equally fluent designing systems and implementing them. Driven by craft: clean, self-documenting, performant code that solves hard problems elegantly. Currently completing an M.S. in Information Security (B.B.A. Computer Science, Summa Cum Laude).

COMPLETE TECHNOLOGY STACK

Languages

C# · JavaScript · TypeScript · Node.js · PHP · VB.NET · VB6 · SQL · T-SQL · PL/SQL · HTML5 · CSS3 · XML · XSLT · XPath · JSON · YAML · PowerShell · Bash · Python · Lua · ShaderLab · HLSL · GLSL · Markdown · Regex

C# / .NET Ecosystem

.NET Framework · .NET Core · .NET 5–10 · ASP.NET Web Forms · ASP.NET MVC · ASP.NET Core · ASP.NET Web API · WCF · SignalR · MAUI · Blazor Server · Blazor WASM · WPF · WinForms · Worker Services · Quartz.NET · Hangfire · LINQ · Entity Framework 6 · EF Core · Dapper · ADO.NET · AutoMapper · MediatR · FluentValidation · Polly · Serilog · Newtonsoft.Json · System.Text.Json · RestSharp · HttpClient · gRPC · Protobuf · GraphQL (HotChocolate) · OData · xUnit · NUnit · MSTest · Moq · FluentAssertions

Frontend Frameworks

React · React Hooks · React Router · Next.js · Angular · AngularJS · Vue.js · Svelte · jQuery · jQuery UI · Backbone.js · Knockout.js · Alpine.js · Htmx · Redux · Zustand · MobX · Tanstack Query · SWR · Axios · Socket.io · Lodash · Moment.js · Day.js · Zod · Formik · React Hook Form · Framer Motion · GSAP · Three.js · D3.js · Chart.js · Recharts · Leaflet

Styling & Design

CSS3 · CSS Grid · Flexbox · CSS Variables · CSS Animations · Tailwind CSS · Bootstrap 3–5 · Material UI · Ant Design · Chakra UI · Radix UI · shadcn/ui · Styled Components · CSS Modules · SASS / SCSS · Less · PostCSS · BEM · Figma · Adobe XD · Sketch · Storybook · Design tokens · Responsive design · Mobile-first · Accessibility (WCAG 2.1)

Build Tools

Webpack · Vite · Parcel · Rollup · esbuild · Babel · ESLint · Prettier · npm · Yarn · pnpm · Grunt · Gulp · MSBuild · Cake · Nx · Turborepo · Lerna

Backend & Architecture

REST APIs · GraphQL · WebSockets · gRPC · WebRTC · SSE · Webhooks · Microservices · Event-driven architecture · CQRS · Event Sourcing · DDD · Hexagonal architecture · Clean architecture · Repository pattern · SOLID · Pub/Sub · Message queues · Saga pattern · Outbox pattern · Circuit breaker · API Gateway · BFF pattern · Feature flags · Rate limiting · API versioning

Databases & Storage

SQL Server 2005–2022 · T-SQL · Stored procedures · Views · Triggers · CTEs · Window functions · Full-text search · SQLite · PostgreSQL · MySQL · MariaDB · Redis · Elasticsearch · OpenSearch · Azure Cosmos DB · Azure Blob Storage · MongoDB (concepts) · Database design · Normalization · Indexing strategies · Query optimization · EF Core migrations

AI / ML & Data Science

LLM integration · RAG pipelines · Vector embeddings · Vector databases · LSH (Locality-Sensitive Hashing) · SimHash · MinHash · Semantic search · Cosine similarity · TF-IDF · BM25 · Tokenization · Chunking strategies · Prompt engineering · Few-shot prompting · Chain-of-thought · Tool use / function calling · OpenAI API · Anthropic API · Azure OpenAI · Semantic Kernel · Genetic algorithms · Neural networks · Gradient descent · Backpropagation · NLP · Phishing detection · Anomaly detection · Backtesting · Technical indicators (RSI, MACD, ADX, VWAP, EMA, Bollinger Bands, ATR) · Position sizing · Monte Carlo simulation

Data & Visualization

D3.js · Force-layout graphs · Graph algorithms · BFS / DFS · A* pathfinding · Canvas API · SVG · WebGL · Real-time data pipelines · Cross-SOR data unification · Bidirectional push/pull sync · CDC · Azure Service Bus · RabbitMQ · ETL concepts · Excel generation (EPPlus) · PDF generation · CSV parsing · XSLT transformation

Security & Infosec

OWASP Top 10 · SQL injection prevention · XSS prevention · CSRF protection · CSP · CORS · HTTPS / TLS · OAuth 2.0 · OpenID Connect · SAML 2.0 · JWT · RBAC · ABAC · Active Directory · LDAP · Windows Authentication · AES encryption · RSA encryption · bcrypt · Argon2 · HMAC · Azure Key Vault · Vulnerability scanning · Pen testing concepts · SAST / DAST · NIST frameworks · CIS benchmarks · Threat modeling · STRIDE · MFA · SIEM concepts · Log analysis

Finance & Trading

Interactive Brokers API (IBKR) · TWS API · IB Gateway · Algorithmic trading · Automated order execution · Market/Limit/Stop/OCO orders · Paper trading · Live trading · Position management · Risk management · Drawdown analysis · Sharpe ratio · Gap trading · Momentum strategies · Mean reversion

Trend following · Backtesting engines · Walk-forward optimization · DSL design · IdiotScript · Candlestick patterns · Support & resistance · Volume analysis · OHLCV data · TradingView · Pine Script

Game Development

Unity · Unity C# scripting · Unity Physics · Unity Animator · Unity Canvas UI · Unity Input System · Unity Profiler · ShaderLab · HLSL · Shader Graph · Compute Shaders · Post-processing · Procedural generation · Perlin noise · Pathfinding (A*) · NavMesh · State machines · ECS (concepts) · Tilemaps · Sprite rendering · Particle systems · Level design · World building · Narrative design · URP / HDRP · Performance optimization

Cloud & Infrastructure

Azure · Azure App Service · Azure Functions · Azure DevOps · Azure Pipelines · Azure Blob Storage · Azure Service Bus · Azure Event Grid · Azure Cosmos DB · Azure SQL · Azure Cache for Redis · Azure API Management · Azure Key Vault · Azure Monitor · Azure Application Insights · Azure Active Directory · Docker · Docker Compose · Kubernetes (concepts) · Terraform (concepts) · Nginx · IIS · Windows Server · Linux (Ubuntu) · CI/CD · GitHub Actions

CMS & Content

WordPress · WordPress plugin dev · WordPress theme dev · WordPress REST API · PHP CMS development · Custom CMS development · SEO optimization · Structured data · Open Graph · Sitemap generation · Page speed optimization · Core Web Vitals · Content strategy · Social media integration

Testing & Quality

Unit testing · Integration testing · E2E testing · TDD · BDD · xUnit · NUnit · MSTest · Moq · FluentAssertions · Bogus · SpecFlow · Playwright (concepts) · Selenium (concepts) · k6 (concepts) · Code coverage · Static analysis · SonarQube (concepts) · QA process management · UAT

Tools & Workflow

Visual Studio 2005–2022 · VS Code · JetBrains Rider · LinqPad · SSMS · Azure Data Studio · Postman · Swagger / OpenAPI · Fiddler · Charles Proxy · Chrome DevTools · Git · GitHub · GitKraken · Beyond Compare · PowerShell ISE · Windows Terminal · WSL2 · FileZilla · WinSCP · Sysinternals

Engineering Practices

Agile / Scrum · Kanban · Sprint planning · Backlog grooming · Technical debt management · Architecture decision records · Self-documenting code · Clean code · DRY / KISS / YAGNI · Code review · Pair programming · Documentation · Patent filing · API design · SDK design · Mentoring · Stakeholder communication · Written Communication · Oral Communication

PROFESSIONAL EXPERIENCE

Senior Software Engineer · Bank of America — Oct 2013 – Present

Chicago, IL

C# · .NET · React · Angular · JavaScript · TypeScript · HTML/CSS3 · SQL Server · Elasticsearch · SignalR · REST APIs · GraphQL · PetaPoco

- **Datastrology** — Designed and built a data-agnostic, multi-threaded force-layout graph visualization engine mapping relationships across complex business entities at enterprise scale. *Patent Pending.*
- **Graphite** — Architected a cross-SOR data unification platform connecting heterogeneous sources (SQL, REST APIs, Elasticsearch, Excel) into a single live document. Bidirectional push/pull sync, no caching, no UI reload.
- **Trident** — Engineered an ML-driven phishing correlation engine identifying and clustering multi-vector threat actors across global infrastructure.
- **Continuous Monitoring** — Architected an enterprise-wide vulnerability identification and remediation platform deployed across global systems.
- **Bonsai Menu** — Invented a novel hierarchical navigation paradigm. *US Patent #US-9933917-B2.*
- **Gamified Training App** — Co-invented an associate training platform with unique gamification mechanics. *US Patent #US-9747636-B2.*
- **GDS** — Designed a responsive UI/UX component library standardized across all GIS projects.
- **INSTINCT / Canary** — Modernized multiple legacy applications delivering significant performance improvements.
- **InfoPass** — Developed a corporate-wide traveler analytics and tracking dashboard.

CMS Developer / Managing Editor · Bag of Games — Dec 2011 – Oct 2013

Helsinki, Finland (Remote)

PHP | WordPress | JavaScript | jQuery | HTML | CSS | SEO

- Rebuilt site frontend in PHP with a modern UI and tighter social media integration.
- Promoted to Managing Editor — led a team of 10 writers, owned all CMS technical operations.
- Authored 90+ articles; developed content strategy optimized for longevity, SEO, and readership growth.

.NET Developer (Contract) · Kimberly-Clark — Dec 2007 – Oct 2010

Neenah, WI

C# | ASP.NET | VB.NET | SQL Server | XML | Active Directory | QA/UAT

- **RADAR X** — Led ground-up C#/SQL Server replacement of a legacy regulatory document management platform.
- Built CSAS (access management), RIM (tag-based CMS), and KC Portal (dynamic internal link portal).
- Managed full QA lifecycle; facilitated user acceptance testing and training sessions.

CORPORATE

CEO · MindAttic Interactive — Est. Jul. 10, 2017

LLC · 5940 S Rainbow Blvd Ste 400 #46782, Las Vegas, NV 89118

President · MindAttic Cares — Est. Sept. 29, 2025

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PERSONAL PROJECTS

Tutor — AI-powered adaptive learning platform (C#/.NET MAUI). Full RAG pipeline with LSH/SimHash concept matching, force-layout knowledge graphs, and an evolving textbook that retroactively layers new knowledge into earlier chapters. Adaptive quizzes, live news injection, user progress tracking. Cross-platform.

IdiotProof — Autonomous AI trading system (C#/.NET 10) with custom DSL (IdiotScript), genetic algorithm optimizer, and neural network weight learning. Dual ML methods self-calibrate per-ticker indicator weights. Live IBKR order execution with comprehensive backtesting. Built on Ahab — a full-featured multi-strategy trading platform with core engine, backtesting, and broker integration — and Gappy, a live gap-trading system with real-time indicator pipeline, modular strategy framework, and GUI dashboard.

GridGame2026 — Original grid-based strategy game in Unity with custom ShaderLab shaders and self-authored world lore. Active development since 2025.

ChatGPT_To_Claude — JavaScript migration tool exporting full ChatGPT conversation history into Claude Projects via the Anthropic API.

EDUCATION

M.S. Information Security · Southern New Hampshire University — In Progress

B.B.A. Computer Science · Southern New Hampshire University · *Summa Cum Laude* — Graduated

PATENTS & INVENTIONS

Hierarchical Navigation System (Bonsai Menu) (US-9933917-B2)
Novel hierarchical data navigation methodology eliminating depth limitations of traditional menus.

Gamified Training Platform (US-9747636-B2)
Associate training application implementing unique gamification mechanics for engagement and retention.

Datastrology — Force-Layout Visualization Engine (Patent Pending)
Data-agnostic multi-threaded force-layout visualization engine for mapping business entity relationships.

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